TEST-02-MATERIALS-TESTING-PROTOCOL

TEST-02: MATERIALS TESTING PROTOCOLS

Carrington Storm Motors — Project AEGIS

MXene • BFRP • Ceramics • CNT-Polymer • Geopolymer • Piezoelectrics • Aerogels • MRF • STF

Document Status: Technical Test Protocol — NIST-Traceable
Version: 1.0 — 15 July 2026
Classification: CONFIDENTIAL — Manufacturing Process Data Redacted in External Distribution
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Reference Standards: IEEE 299, MIL-STD-188-125, ASTM D149, ASTM D257, ASTM D4935, ASTM D3039, IEC 60243, ASTM C39, ISO 178

PROTOCOL 1: MXene $Ti_3C_2T_x$ SHIELDING EFFECTIVENESS

1.1 Reference Standards

- **IEEE 299-2006:** Standard Method for Measuring the Effectiveness of Electromagnetic Shielding Enclosures
- **MIL-STD-188-125-1:** High-Altitude Electromagnetic Pulse (HEMP) Protection for Ground-Based C4I Facilities
- **ASTM D4935-18:** Standard Test Method for Measuring the Electromagnetic Shielding Effectiveness of Planar Materials

1.2 Test Specimen Preparation

Parameter	Specification	Rationale
MXene formulation	$Ti_3C_2T_x$, synthesized via CSM electrochemical etching process	Consistent surface termination chemistry verified by XPS
Substrate	BFRP/Elium composite panel, 100 × 100 × 3 mm	Representative of final product substrate
MXene layer thickness	45 ± 2 µm (target); 20, 30, 45, 60, 90 µm test matrix for optimization	Verify SE vs. thickness relationship; identify diminishing returns threshold
Coating method	Multi-nozzle spray with in-situ spectroscopic thickness feedback	Production-representative method
Surface termination (XPS)	F <5 at%, O >60 at%, OH <20 at%, Cl <5 at%	Fluoride-free process target; excess F degrades SE
Number of test specimens	5 per thickness (25 total for thickness matrix); additional 10 for durability/post-exposure testing	Statistical significance; n=5 gives detectable effect size of 1 standard deviation at $\alpha=0.05$, $\beta=0.20$
Conditioning	24 hours at 23 ± 2°C, 50 ± 5% RH prior to testing	ASTM D4935 standard conditioning

1.3 Test Equipment

Equipment	Make/Model	Calibration	Uncertainty
TEM Cell	Custom CSM design (2 m × 1 m × 1 m working volume)	IEEE 299 field uniformity verified annually	±1.5 dB SE
Vector Network Analyzer	Keysight N5227A PNA, 10 MHz–67 GHz	Keysight calibration lab, annual	±0.1 dB amplitude

Field Probes (E-field)	AR FL7006, 10 kHz–6 GHz	NIST-traceable cal, annual	±0.5 dB (k=2)
Field Probes (H-field)	Beehive 100D, 10 kHz–3 GHz	NIST-traceable cal, annual	±0.5 dB (k=2)
Profilometer	Bruker Dektak XT	NIST-traceable step height standard, quarterly	±0.1 μm
XPS	Thermo Scientific K-Alpha	Sputter-cleaned Au/Ag/Cu reference, monthly	±0.1 at%
4-Point Probe	Jandel RM3000+	NIST SRM thin-film resistivity, semi-annual	±2% of reading

1.4 Test Procedure

1. **Pre-Test:** Measure MXene thickness at 9 points (3×3 grid) via profilometer; record XPS surface termination; measure DC sheet resistance via 4-point probe at 5 points; photograph specimen
2. **TEM Cell Calibration:** Measure empty cell S-parameters; verify field uniformity within ±2 dB across specimen mounting area
3. **Reference Measurement:** Mount uncoated BFRP substrate in TEM cell; sweep 1 kHz–10 GHz, 1,001 points; record S21 (transmission) as reference
4. **MXene Specimen Measurement:** Mount MXene-coated specimen; identical sweep parameters; record S21
5. **SE Calculation:** SE(dB) = S21_reference(dB) − S21_specimen(dB)
6. **Repeat:** 3 sweeps per specimen; average; report standard deviation
7. **Post-Test:** Re-measure MXene thickness; check for delamination or cracking; photograph

1.5 Pass/Fail Criteria

Parameter	Criterion	Accept/Reject
SE at 1 kHz	>80 dB	ACCEPT
SE at 10 MHz	>85 dB	ACCEPT
SE at 1 GHz	>90 dB	ACCEPT
SE at 10 GHz	>85 dB (roll-off acceptable above 8 GHz)	ACCEPT
Full-band minimum SE	>85 dB anywhere in 1 kHz–10 GHz	ACCEPT (production); >90 dB (aerospace/defense grade)
Thickness tolerance	45 ± 2 μm	ACCEPT; if >50 μm, investigate spray process; if <40 μm, reject
Adhesion (visual)	No delamination, cracking, or flaking after TEM cell test	ACCEPT
DC resistance between tiles (discontinuous pattern)	>10^8 Ω between any pair of adjacent tiles	ACCEPT (confirms DC isolation)

1.6 Durability Testing (Post-TEM Cell)

Test	Method	Duration/Cycles	Post-Test SE Degradation Limit
Salt Spray	ASTM B117, 5% NaCl, 35°C	500 hours	<3 dB degradation
UV Exposure	ASTM G154, Cycle 1 (UVA-340, 8h UV at 60°C / 4h condensation at 50°C)	2,000 hours	<5 dB degradation
Thermal Cycling	−40°C to +85°C, 5°C/min ramp, 1 hour dwell	100 cycles	<2 dB degradation
Humidity	85°C / 85% RH	1,000 hours	<2 dB degradation
Abrasion	Taber abraser, CS-10 wheels, 500 g load	1,000 cycles	<5 dB degradation (post-abrasion area only)
Adhesion (Cross-Hatch)	ASTM D3359, Method B	1 test per specimen	>4B rating (no more than 5% area removed)

Cost per Complete MXene Test Suite: $5,200 (materials + CSM labor) or $8,500 (third-party salt

spray/UV)

PROTOCOL 2: BFRP DIELECTRIC STRENGTH

2.1 Reference Standards

- **ASTM D149-20:** Standard Test Method for Dielectric Breakdown Voltage and Dielectric Strength of Solid Electrical Insulating Materials at Commercial Power Frequencies
- **IEC 60243-1:** Electrical Strength of Insulating Materials — Tests at Power Frequencies

2.2 Test Specimens

Parameter	Specification
Material	BFRP/Elium composite, 60% basalt fiber volume fraction
Specimen dimensions	100 × 100 × 3 mm (standard); 1, 2, 3, 5, 10 mm thickness test matrix
Number of specimens	5 per thickness (25 total); n=5 per IEC 60243 minimum
Electrode configuration	Type 2 electrodes — 25 mm diameter cylindrical electrodes with 3 mm edge radius per ASTM D149
Conditioning	48 hours at 23 ± 2°C, 50 ± 5% RH; optional: 24 hours at 50°C to drive off moisture
Additional conditioning	Submersion in deionized water for 24 hours (per IEC 60243 for moisture sensitivity) — separate specimen set

2.3 Test Equipment

Equipment	Specification
High-Voltage Supply	0–100 kV AC, 60 Hz, 5 kVA minimum; ramp rate 500 V/s per ASTM D149 (short-time test)
Voltage Measurement	Precision voltage divider, ±1% accuracy
Current Detection	Circuit breaker set to trip at 10 mA short-circuit current
Test Medium	Transformer oil (ASTM D3487 Type I) to prevent flashover around specimen edges
Electrodes	Brass, Type 2 geometry per ASTM D149, polished to 0.8 μm Ra surface finish

2.4 Test Procedure

1. Immerse specimen in transformer oil bath (prevents surface flashover — measures true bulk dielectric strength)
2. Apply voltage at ramp rate 500 V/s (short-time method) until breakdown (current exceeds 10 mA)
3. Record breakdown voltage, specimen thickness at breakdown point, and time to breakdown
4. Calculate dielectric strength: $E_{bd} = V_{bd} / d$ (kV/mm), where d is measured thickness at puncture point
5. Repeat for all 5 specimens per thickness; report mean, standard deviation, and minimum value

2.5 Pass/Fail Criteria

Parameter	Criterion
Dielectric strength (3 mm, dry)	>15 kV/mm
Dielectric strength (3 mm, 24h water immersion)	>10 kV/mm (moisture degrades 30–40% — acceptable for indoor applications)
Coefficient of variation (CV) across 5 specimens	<15%

PROTOCOL 3: CNT-POLYMER CONDUCTIVITY (EMBARK FOOTWEAR SOLE)

3.1 Reference Standard

- **ASTM D257-14(2021):** Standard Test Methods for DC Resistance or Conductance of Insulating Materials

3.2 Target Resistivity Window

Parameter	Target	Rationale
Surface resistivity (ρs)	10^6–10^8 Ω/sq	ESD-safe range (10^5–10^{11} Ω/sq per ANSI/ESD S20.20); lower bound ensures static dissipation; upper bound ensures GIC isolation (>10^8 Ω prevents power-frequency current flow)
Volume resistivity (ρv)	10^4–10^6 Ω·cm	Through-thickness resistivity for sole material
Resistance (point-to-point)	10^5–10^7 Ω	Point-to-point per ANSI/ESD STM11.13

PROTOCOL 4: CERAMIC BEARING DIELECTRIC ISOLATION (Si_3N_4)

4.1 Reference Standard

- **IEC 60243-1** (as modified for ceramic bearing geometry)
- **ASTM F2094:** Standard Specification for Silicon Nitride Bearing Balls

4.2 Test Setup

- Si_3N_4 bearing ball, 10–25 mm diameter, placed between parallel plate electrodes
- Applied voltage: DC, 500 V/s ramp, in transformer oil
- Measure breakdown voltage across bearing diameter
- Dielectric strength = V_bd / ball diameter

4.3 Pass/Fail Criteria

Parameter	Criterion
Dielectric strength	>20 kV/mm (Si_3N_4 bulk material property exceeds 25 kV/mm; target >80% of bulk)
Surface flashover (if occurs before bulk breakdown)	>15 kV/mm acceptable; indicates surface contamination — clean and re-test
Fatigue life	>3× equivalent steel bearing (per ASTM F2094 rolling contact fatigue test — separate long-duration test)

PROTOCOL 5: GEOPOLYMER CONCRETE RESISTIVITY

5.1 Reference Standards

- **ASTM C39:** Compressive Strength of Cylindrical Concrete Specimens
- **ASTM C1202:** Electrical Indication of Concrete's Ability to Resist Chloride Ion Penetration (Rapid Chloride Permeability Test — RCPT) — modified to measure bulk resistivity
- **ASTM C1760:** Bulk Electrical Conductivity of Hardened Concrete

5.2 Test Specimens

Parameter	Specification
Mix design	Class C fly ash + ground granulated blast furnace slag (GGBFS), alkali-activated with $NaOH/Na_2SiO_3$ solution; aggregate per ASTM C33

Specimen geometry	Φ100 × 200 mm cylinders (compressive); Φ100 × 50 mm discs (resistivity)
Curing	28 days at 23 ± 2°C, >95% RH (sealed curing — geopolymer requires moisture retention)
Conditioning for resistivity	Saturated surface-dry (SSD) condition — 48 hours water immersion, surface-dried prior to test

5.3 Resistivity Test Procedure (ASTM C1760)

1. Place SSD specimen between two stainless steel plate electrodes with conductive gel couplant
2. Apply AC voltage (10 V, 60 Hz — avoids polarization effects of DC measurement)
3. Measure current (I) and phase angle (θ)
4. Calculate resistivity: $\rho = (V \times A) / (I \times L)$, where A = electrode contact area, L = specimen thickness
5. Correct for temperature to 23°C standard: $\rho_{23} = \rho_T \times \exp[Ea/R \times (1/T - 1/296)]$, where Ea = 30 kJ/mol for geopolymer

5.4 Pass/Fail Criteria

Parameter	Criterion
Compressive strength (28-day)	>60 MPa for structural applications; >40 MPa for non-structural
Bulk resistivity (saturated)	>10^8 Ω·m (two orders of magnitude above conventional Portland cement concrete, ~10^6 Ω·m)
Resistivity (oven-dry)	>10^{10} Ω·m (confirms moisture is primary conductivity mechanism — acceptable)
Rapid Chloride Permeability (ASTM C1202, modified)	<100 coulombs (negligible chloride ion penetrability; typical geopolymer achieves <500 C vs. >4,000 C for OPC)

PROTOCOL 6: YInMn BLUE / $CsPbBr_3$ PEROVSKITE — DIELECTRIC PERMITTIVITY

6.1 Reference Standards

- **ASTM D150-18:** Standard Test Methods for AC Loss Characteristics and Permittivity (Dielectric Constant) of Solid Electrical Insulation
- **IEC 60250:** Recommended Methods for the Determination of the Permittivity and Dielectric Dissipation Factor of Electrical Insulating Materials at Power, Audio, and Radio Frequencies

6.2 Test Setup

Parameter	Specification
Frequency range	100 Hz – 1 MHz (for building material applications); 1 MHz – 1 GHz (for EMI coating applications)
Specimen	YInMn Blue pigmented coating on BFRP substrate, 100 × 100 mm, coating thickness 100–200 μm
Electrode configuration	Guarded electrode per ASTM D150 to eliminate fringe field effects
Measurement	Capacitance bridge (100 Hz–1 MHz) or impedance analyzer (1 MHz–1 GHz)

6.3 Pass/Fail Criteria

Parameter	Criterion
Relative permittivity (ε_r) at 1 kHz	<10 (target; typical ceramic pigments ε_r=5–15; must be < Aegis-C panel target for GIC-free capacitive coupling)
Dissipation factor (tan δ) at 1 kHz	<0.02

Dielectric strength (ASTM D149, through-thickness of pigmented coating)	>10 kV/mm

PROTOCOL 7: PIEZOELECTRIC ($KNbO_3$-$BaTiO_3$) CHARACTERIZATION

7.1 Reference Standard

- **IEEE/ANSI 176-1987:** IEEE Standard on Piezoelectricity (resonance-antiresonance method)

7.2 Key Parameters Measured

Parameter	Symbol	Target	Test Method
Piezoelectric charge coefficient	d_{33}	>250 pC/N (target 285 per CSMFAB BOM-REGISTRY)	Berlincourt d_{33} meter (quasi-static) or laser interferometer (dynamic)
Electromechanical coupling coefficient	k_p	>0.45 (radial mode)	IEEE 176 resonance-antiresonance (impedance analyzer sweep, identify f_r and f_a)
Curie temperature	Tc	>300°C	Capacitance vs. temperature sweep (2°C/min, 1 kHz); Tc = temperature at peak permittivity
Mechanical quality factor	Qm	>500	IEEE 176 — 3 dB bandwidth of resonance peak
Dielectric constant (free)	$\varepsilon^T_{33}/\varepsilon_0$	800–1,200	Capacitance at 1 kHz, well below resonance (typically 100 Hz)
Dissipation factor (free)	tan δ	<0.02 at 1 kHz	Same measurement as ε^T_{33}
Density	ρ	>95% theoretical	Archimedes method

7.3 Poling Procedure

- Temperature: 120°C (above Tc/3, below Tc to avoid depoling)
- DC electric field: 3 kV/mm applied for 30 minutes
- Cool to room temperature under field (field-cooling prevents immediate depoling)
- Age for 24 hours before measurement (piezoelectric properties stabilize after poling)

PROTOCOL 8: POLYIMIDE-SILICA AEROGEL (THERMAL CONDUCTIVITY)

8.1 Reference Standard

- **ASTM C518-21:** Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus

8.2 Test Specimens

Parameter	Specification
Dimensions	300 × 300 × 25 mm (standard for heat flow meter)
Density	180 ± 10 kg/m³
Conditioning	24 hours at 23 ± 2°C, 50 ± 5% RH
Temperature gradient	20°C (mean specimen temperature 25°C, ΔT = 20°C,

	cold side 15°C, hot side 35°C)

8.3 Pass/Fail Criteria

Parameter	Criterion
Thermal conductivity (λ) at 25°C mean temperature	<0.015 W/m·K (target 0.010 per BOM-REGISTRY)
Compressive strength	>0.8 MPa at 10% strain (ASTM C165)
Service temperature (continuous)	Up to 650°C (per polyimide-silica aerogel specification; verified by TGA)

PROTOCOL 9: MRF-140CG MAGNETORHEOLOGICAL FLUID — RHEOLOGICAL CHARACTERIZATION

9.1 Reference Standards

- No single ASTM standard for MR fluid — test methods adapted from:
  - ASTM D2196 (rotational viscometry)
  - Custom magnetorheological test fixture

9.2 Test Equipment

Equipment	Specification
Rheometer	Anton Paar MCR 302 with MRD (magnetorheological device) accessory — provides controlled magnetic field up to 1 T
Measurement geometry	Parallel plate, 20 mm diameter, 0.5 mm gap

9.3 Key Parameters

Parameter	Symbol	Target	Test Conditions
Off-state viscosity	η_0	<0.5 Pa·s (target 0.28 per BOM-REGISTRY)	Shear rate 10 s^{-1}, B=0 T
Yield stress (on-state)	τy	>60 kPa (target 80 per BOM)	B=250 kA/m (≈0.31 T in MRD gap), shear rate 0.1 s^{-1}
Response time	t_response	<10 ms	Step change in B field, measure time to 90% of steady-state τ
Sedimentation stability	—	<5% by volume after 30 days static	Graduated cylinder observation
Temperature range	—	−40°C to +130°C operational	Rheometry at temperature extremes; verify τy within 20% of room-temperature value

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PROTOCOL 10: SHEAR-THICKENING FLUID (STF) — THORAX-CALM SEATBELT

10.1 Test Methods

Parameter	Method	Target
Critical shear rate (onset of thickening)	Rotational rheometry, shear rate sweep 0.1–1,000 s^{-1}	10–100 s^{-1} (tuned to collision-relevant shear rate)
Peak viscosity (thickened state)	Same sweep — measure maximum η	>10 Pa·s (4 kPa·s at 1,000 s^{-1} per BOM-REGISTRY)

Reversibility	Cycle shear rate 0.1→1,000→0.1 s^{-1}, 10 cycles	Viscosity returns to baseline within ±5% (confirms STF is reusable after impact)
Temperature stability	Repeat rheometry at −20°C, 23°C, 60°C	Critical shear rate shift <20%, peak viscosity within 30% of room-temperature value
Long-term stability	Measure weekly for 12 months (sealed container, ambient)	<10% change in critical shear rate and peak viscosity

BATCH TESTING — PRODUCTION QUALITY CONTROL SAMPLING

Sampling Plan (ANSI/ASQ Z1.4, General Inspection Level II, Normal)

Material	Lot Size	Sample Size	AQL (Acceptance Quality Limit)	Critical Defect (Major)
MXene (per 500 g batch)	N/A (batch processing)	1 specimen per batch for SE; 1 per 10 batches for full suite	0.65%	SE <85 dB at any frequency; thickness out of 45±5 μm
BFRP (per pultrusion run, ~1,000 m)	1,000 m	5 specimens per run	1.0%	Tensile <1,000 MPa; dielectric <12 kV/mm; void >3%
Ceramic bearings (Si_3N_4, per 1,000 pcs)	1,000 pcs	32 pcs (AQL 1.0, Normal)	1.0%	Surface defect visible at 5× magnification; out-of-round >2 μm
Geopolymer (per truckload, ~8 m^3)	8 m^3	3 cylinders per truck	2.5% (minor)	Compressive <50 MPa at 7-day (early strength indicator)
Piezoelectric elements (per 100 pcs)	100 pcs	13 pcs (AQL 2.5, Normal)	2.5%	d_{33} <200 pC/N; tan δ >0.05; visible crack
Aerogel panels (per 50 pcs)	50 pcs	8 pcs (AQL 2.5)	2.5%	λ >0.020 W/m·K; compressive <0.5 MPa; density >220 kg/m^3

EQUIPMENT CALIBRATION SCHEDULE — MATERIALS LAB

Equipment	Calibration Interval	Reference Standard	Calibration Provider	Cost per Cal
TEM Cell (field uniformity)	12 months	IEEE 299 mapping with NIST-calibrated probe	In-house or NTS	$4,500
VNA (Keysight N5227A)	12 months	Keysight calibration kit (NIST-traceable)	Keysight service	$3,200
Field Probes	12 months	NIST-traceable calibration lab	AR RF/Microwave Instrumentation	$1,800/probe
Universal Test Machine (Instron)	12 months	ASTM E4 Class 1 load verification	Instron service	$2,100
Profilometer	3 months	NIST step height standard	In-house verification	$0 (in-house)
XPS	Monthly (binding energy)	Sputter-cleaned Au 4f7/2 (84.0 eV), Ag 3d5/2 (368.3 eV), Cu 2p3/2 (932.7 eV)	In-house standard	$0 (in-house)
4-Point Probe	6 months	NIST SRM thin-film resistivity standard	In-house verification	$0 (in-house)
Environmental		NIST-calibrated	In-house or third-	

Chamber (T/RH sensor)	6 months	thermocouple, chilled mirror hygrometer	party	$800
Rheometer (Anton Paar)	12 months	Standard viscosity oil (NIST-traceable)	Anton Paar service	$2,500
High-Voltage Supply + Divider	12 months	Precision voltage divider calibration	Calibration lab (ISO 17025)	$1,500

"Materials testing is the root of the tree. If the MXene batch has incorrect surface termination, every Aegis-C panel made from it will underperform. If the BFRP pultrusion has excessive voids, every Phantom frame will be weaker than designed. The materials testing protocols in this document are designed to catch those root-cause failures before a single product is assembled. Because it is far cheaper to reject a bad batch of MXene than to recall 500 Aegis Homes." — Mork, Test Engineer

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Document generated 2026-07-15. All test methods are NIST-traceable through the calibration chain described. Equipment procurement status indicated inline. Test costs are estimated at current (2026) rates for materials and labor. For third-party testing, quotes should be obtained at time of testing — costs shown are budgetary estimates ±25%.